

Technical Advisory Report

Project Innovate 2025-2026

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Summary

This Technical Advisory Report outlines the **proposed development plan** for our selected project within the Project Innovate 2025–2026 course.

In this report we will be describing our selected project idea, the reason why we chose it and present the main technical considerations involved with it, including:

- Robot design structure
- Hardware components to consider
- Software and control systems recommended
- The chosen implementation strategy
- Expose the overall risks of the project

Multiple alternatives will be presented, alongside a studied and recommended implementation plan for the project, based on practicality, complexity, required components and potential risks.

In this way, this document has as purpose to support the client with decisions regarding the project final expectation product by identifying the **most realistic and effective development options**.

Introduction



This project has as the **main objective** to develop a creative and practical project that **demonstrates technical knowledge, teamwork, and problem-solving skills**, therefore, our team selected the concept of an interactive campus robot that combines mobility, user engagement, and facial recognition within a single system.

The proposed robot is intended to **move safely** around the campus environment using a stable tracked-wheel design and basic navigation systems. When activated through a selected trigger method, the robot will temporarily stop, **detect the closest user's face** through an integrated camera, **analyse facial expressions** using emotion-recognition software, and **provide an appropriate response** through a selected output method such as a screen display or audio prompt.

We have chosen this project because it combines several important areas within the field of Information Technology, including **robotics, embedded systems, computer vision, programming, and user experience design**.

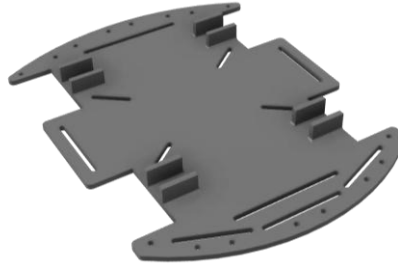
Given that feasibility, complexity, cost, and reliability of the final product are all in question for the development of this system, **this Technical Advisory Report is relevant for all the development options** we have going forward, comparing advantages and disadvantages, and **provide justified recommendations** for the most practical solutions.

Robot Design & Structure

The **physical design** of the robot is a critical factor in **ensuring stability, mobility**, and the **successful integration** of all required hardware and software components.

Given into account that the robot is to operate within a campus environment, **the structure must be practical, durable**, and **capable of moving safely** through indoor spaces such as corridors, open areas and common walkways.

Base Structure



The proposed structure should consist of a **compact rectangular chassis** with the recommended dimensions below to ensure sufficient internal space to house all electronic component securely.

DIMENSION	RECOMMENDED SIZE
LENGTH	35 – 45 cm
WIDTH	28 – 35 cm
HEIGHT OF BASE	12 – 18 cm
TOTAL HEIGHT (WITH MAST + CAMERA)	70 – 100 cm

The **external design** should prioritise accessibility for maintenance, allowing batteries, wiring, and controllers to be reached easily when required.

Protective casing (3D printed for reduced cost) should also be considered to shield sensitive components from accidental contact during operation.

Facial Recognition System

To improve effectiveness, we have opted to incorporate an **elevated camera 3D-printed tube mast** mounted above the main chassis (alternatively a aluminium/PVC tube can be considered).



The camera will be positioned at an approximate height of **85 cm from floor level**, allowing clearer facial capture for standing users while maintaining a compact overall design. This solution avoids the

added mechanical complexity of a retractable lifting camera system while still achieving an appropriate viewing angle.

Besides this, the camera mount will be supported by a **lightweight 3D-printed bracket** connected to a **servo motor** capable of controlling vertical tilt movement ranging approximately:

45° degrees (*15 ° downwards and 30 degrees upwards*)

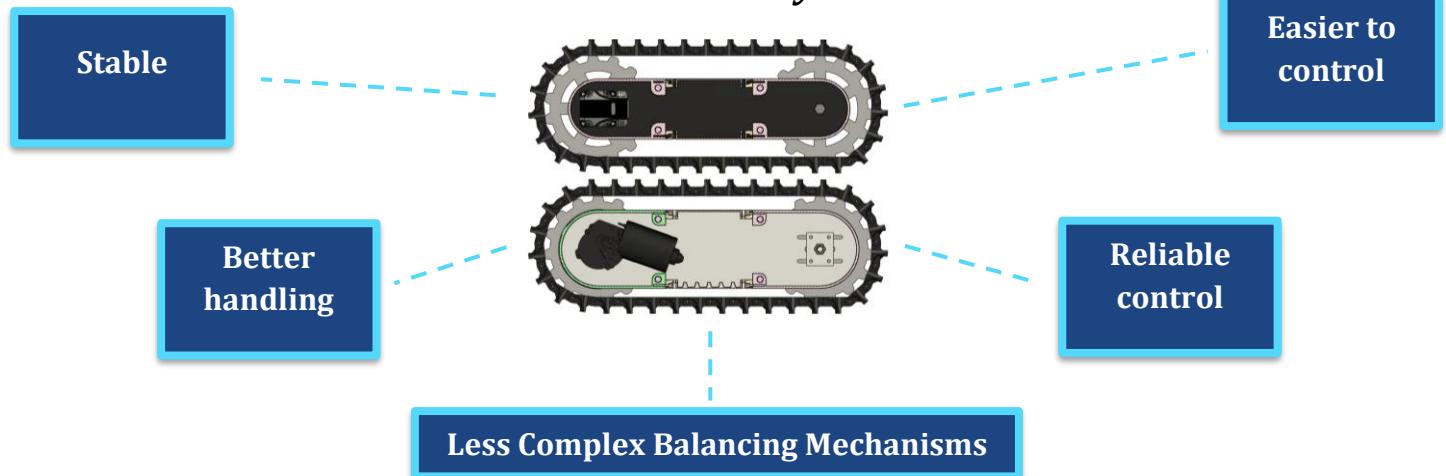
Which will allow the robot to adjust the camera angle to better frame users of different heights and improve image capture reliability during interaction.

CAPTURE TYPE	RANGE
CLOSE/SEATED USERS	~150 cm
TALL STANDING USERS	~190 cm

Mobility

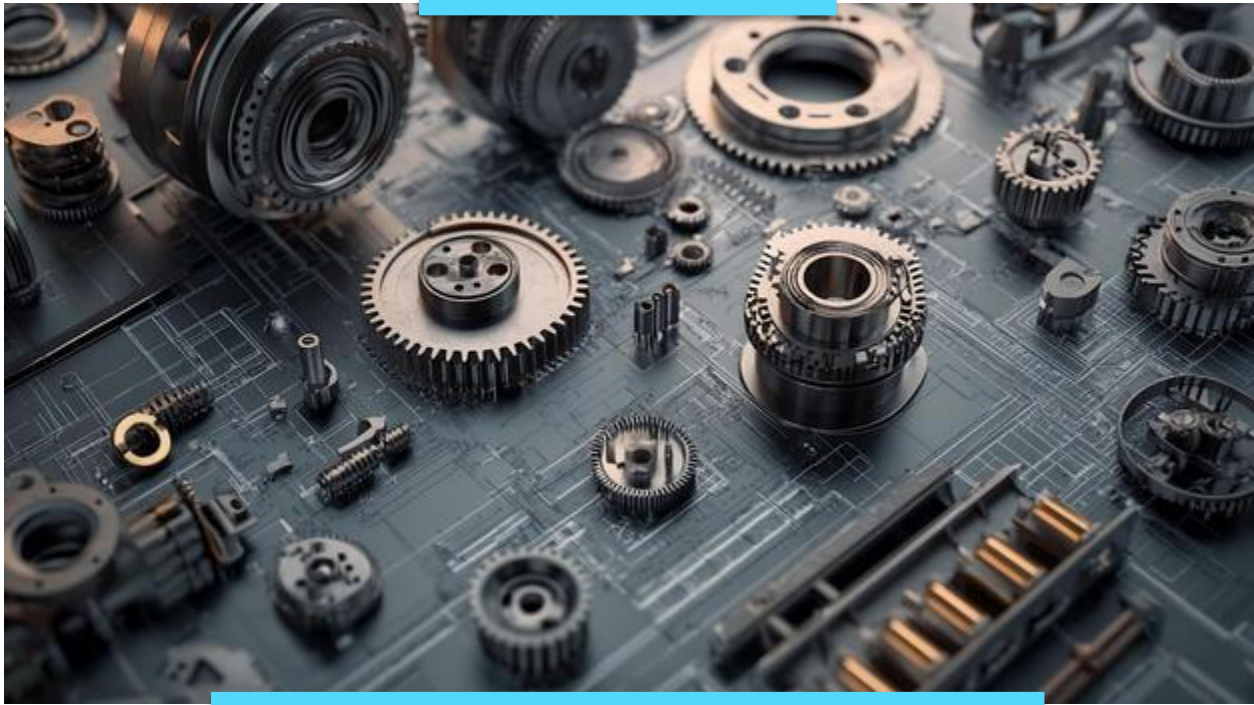
After reviewing several mobility options, we identified the most suitable solution for this project:

Tracked-Wheel System



This will allow the robot to have a **low centre of gravity**, extremely important since it will be carrying sensitive components such as a camera, batteries, processing units and possibly a display screen, which with such stable base will ensure we **can improve** both **safety** and **performance**, while also simplifying the design process.

Hardware Components

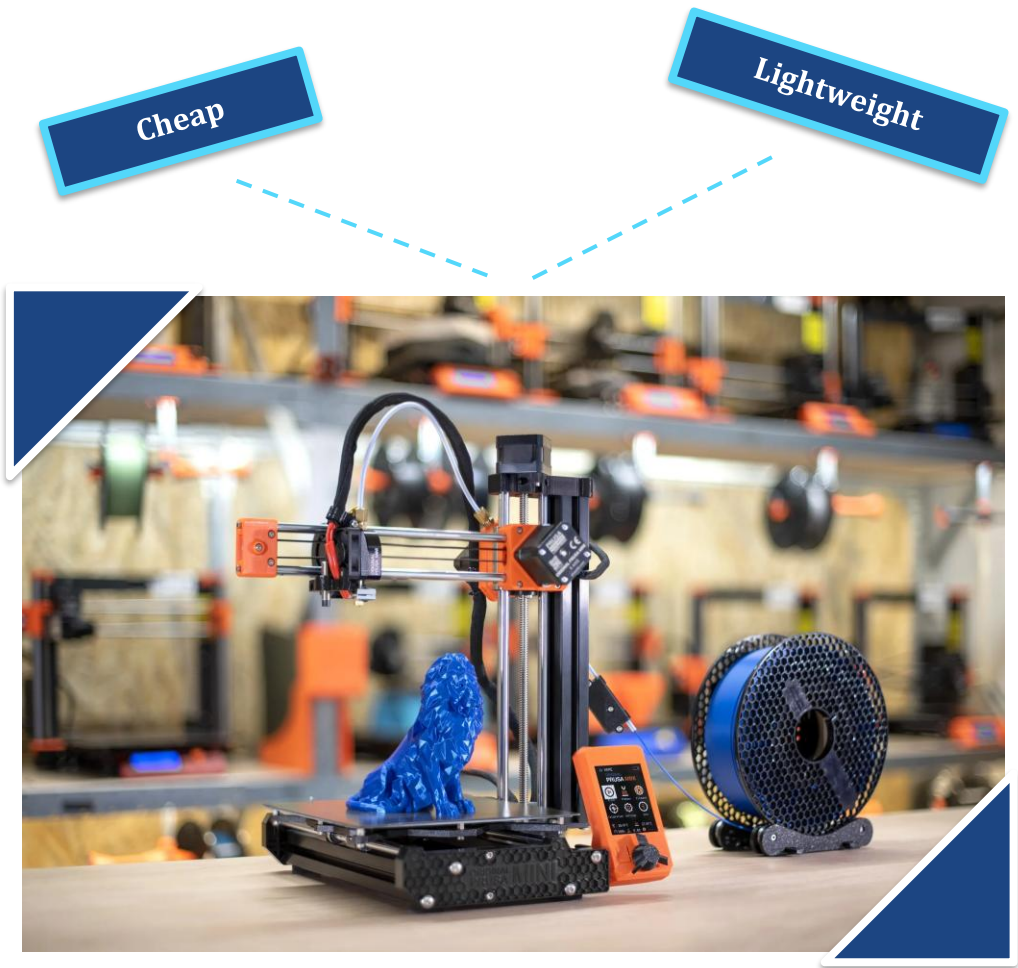


The table below outlines the **primary hardware required** for the recommended implementation, together with their intended functions and justification for selection.

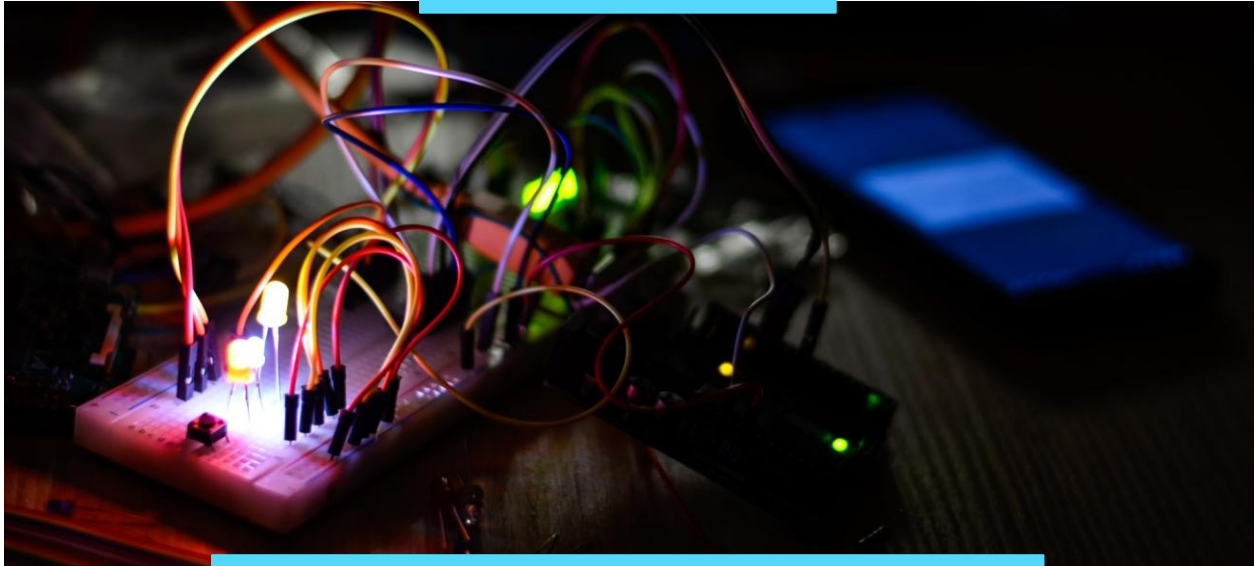
COMPONENT	RECOMMENDED OPTION	FUNCTION	REASON FOR SELECTION
MAIN PROCESSOR	Raspberry Pi 4	Runs Python OpenCV Facial Recognition System Logic	Suitable processing power for intelligent tasks
MICROCONTROLLER	Arduino Uno/ Nano	Controls motors Sensors Real-time hardware actions	Reliable and easy hardware control
TRIGGER METHOD	Push button	Trigger user interaction	Easy to implement and less possibility of missed trigger
MOBILITY SYSTEM	Dual DC Motor + Tracks	Drives robot movement	Stable and simple tracked motion
MOTOR DRIVER	L298N / BTS7960	Controls motor speed and direction	Required interface between motors and controller
CAMERA	UBS Webcam	Captures facial image for analysis	Easy integration with Python and OpenCV
CAMERA SERVO	SG90 / MG996R	Tilts camera upward/downward	Improves user framing
OUTPUT	Speaker with 85dB (2x)	Voice response	Clear user feedback
OBSTACLE SENSORS	Ultrasonic sensors (HC-SR04)	Detect nearby objects	Supports safe navigation

COMPONENT	RECOMMENDED OPTION	FUNCTION	REASON FOR SELECTION
TRIGGER INPUT	Push Button	Activates interaction mode	Simple and reliable
POWER SUPPLY (LOGIC)	Power Bank (5V USB)	Power Raspberry Pi	Stable regulated supply
POWER SUPPLY (MOTORS)	Rechargeable battery pack	Power motors separately	Prevents voltage drops
CHASSIS	Tacked robot base	Supports all components	Stable movement platform

Other components like protective casing, camera mast, bracket and any other holder/support the robot might need, **is recommended to be 3D-printed** for cost reduction and lightweight variability which will less affect robot performance.



Software & Control Systems



The software and control systems of the robot are responsible for **coordinating the movements, user interactions, facial recognition capabilities, and overall decision making.**

Control Systems

We have opted for a **dual-controller architecture**, where the **Raspberry Pi** handles advanced processing tasks and the **Arduino** manages real-time hardware control.

Communication between them will be ensured by a **serial USB connection**, allowing commands and status messages to be exchanged efficiently, which then can be properly handled during programming.

Software Systems



It will be mostly **developed using Python** together with the **OpenCV library** for camera access and face detection.

For control of the hardware we will be using **Arduino IDE**, which works on **C/C++** and where we will be making use of **Serial Communication** for data exchange between Raspberry Pi and Arduino (please inspect the table under the “Communication Commands” for further understanding).

Once a face is detected, we can make use of the **FER or DeepFace libraries** to classify facial expressions into categories which will be first developed as *only* 4 simpler sets at first:

Happy – Easier to detect because of a smile

Neutral – Common baseline expression

Sad – Distinct downward expression

Surprised – Strong visual cues with open eyes and mouth

Upon these emotions are developed and is still inside of our timeframe for the project, we can **further expand** those emotions for the more complex ones to detect, like:

Angry – Could be easily mixed with sad

Fearful – More complex classification

Disgusted – Could be mixed with sadness or anger

Confused – Requires contextual interpretation

Communication Commands

RASPBERRY PI SENDS	ARDUINO SENDS BACK
START	Button Press
STOP	Obstacle Detected
ANALYSE	Movement Complete
RESUME	Ready

Robot Behaviour Workflow

SYSTEM STATE	DESCRIPTION
MOVING	Robot navigates environment
TRIGGERED	User activates interaction mode
STOPPED	Robot safely pauses
SCANNING	Camera searches for face
ANALYSING	Emotion is processed
RESPONDING	Emoji / message displayed
RESUME MOVEMENT	Robot continues movement

Project Design Alternatives



Several technical alternatives were considered during the planning stage of this project. Each option was evaluated based on **implementation difficulty**, **cost**, **reliability**, **user experience**, and **suitability** for the project timeframe. The following comparisons summarise the most relevant design choices and support the recommended final solution.

The following tables will have a final column by the name “R” that stands for Recommendation and proceeds with the following colour scheme:

■ Recommended | ■ Secondary Option | ■ Future Upgrade | ■ Not Recommended

Mobility System

OPTION	DIFFICULTY	REQUIRED COMPONENTS / LIBRARIES	ADVANTAGES	DISADVANTAGES	RISKS	R
TRACKED WHEELS	Easy	Tank chassis	Stable, Strong traction, Simple control	Slightly larger base	Minimal	■
STANDARD WHEELS	Easy	Wheeled chassis	Cheap, Fast	Less traction, Lower stability	Pavement has higher influence	■
LEGGED ROBOT	Very Hard	Legged chassis	Advanced appearance	Complex balance, Harder to code	High failure rate	■

Navigation Strategy

OPTION	DIFFICULTY	REQUIRED COMPONENTS / LIBRARIES	ADVANTAGES	DISADVANTAGES	RISKS	R
RANDOM MOVEMENT + OBSTACLE AVOIDANCE	Easy	Ultrasonic sensors, Motor drivers	Simple, Reliable	Not “smart” navigation	Low	
LINE FOLLOWING	Medium	IR (InfraRed) sensors, Marked paths	Predictable movement	Requires prepared environment	Limited flexibility	
MAPPING / SLAM NAVIGATION	Very Hard	LiDAR, Advanced sensors	Highly advanced	Extremely complex	High failure rate	

Trigger Method

OPTION	DIFFICULTY	REQUIRED COMPONENTS / LIBRARIES	ADVANTAGES	DISADVANTAGES	RISKS	R
BUTTON TRIGGER	Very easy	Push button	Reliable, Simple, Fast response	Requires physical interaction	Minimal	
VOICE KEYWORD DETECTION	Medium	Microphone, Python (Speechrecognition / Porcupine)	Hands-free, More interactive	Struggles in noisy environments	False triggers, No detection	
FULL VOICE RECOGNITION	Hard	Microphone, Cloud / local speech API's	Very advanced interaction	Complex, Slow, Unreliable offline	High failure rate	

Camera System

OPTION	DIFFICULTY	REQUIRED COMPONENTS / LIBRARIES	ADVANTAGES	DISADVANTAGES	RISKS	R
FIXED MAST CAMERA	Very easy	3D printed mast, 3D printed camera support	Reliable, Simple, Good height	Static position	Face might not be captured	
MAST WITH TILT CAMERA	Medium	3D printed mast, 3D printed bracket, Servo motor	Better framing, Adaptive angle	Extra wiring and mechanics	Minimal	
RAISING CAMERA MECHANISM	Very Hard	Scissor lift or telescoping tube	Impressive integration	Complex, Fragile	High failure rate	

Output Method

OPTION	DIFFICULTY	REQUIRED COMPONENTS / LIBRARIES	ADVANTAGES	DISADVANTAGES	RISKS	R
SCREEN EMOJI / TEXT	Easy	LCD, OLED, HDMI screen, Python	Simple, Visual, Reliable	Less expressive	Minimal	
VOICE RESPONSE	Medium	Speaker, Python TTS (text-to-screen) like pyttsx3	More natural interaction	Setup complexity	Minimal	
FULL CHATBOT	Very Hard	Microphone, Speaker, API's	Very impressive	Overkill for a project	High failure rate	

Interaction Style

OPTION	DIFFICULTY	REQUIRED COMPONENTS / LIBRARIES	ADVANTAGES	DISADVANTAGES	RISKS	R
BASIC FUNCTIONALITY	Very easy	Screen, Button	Reliable	Less engaging	Minimal	
PERSONALITY RESPONSES	Medium	Speaker and / or screen	More engaging	Extra design work	Minimal	
ADAPTIVE AI PERSONALITY	Very Hard	Advanced AI	Very impressive	Too complex	High failure rate	

System Architecture

OPTION	DIFFICULTY	REQUIRED COMPONENTS / LIBRARIES	ADVANTAGES	DISADVANTAGES	RISKS	R
RASPBERRY PI + ARDUINO	Medium	Raspberry Pi, Arduino, Serial comms	Clear separation of tasks	Requires integration	Moderate	
ONLY RASPBERRY PI	Medium	Raspberry Pi	Simpler wiring	Harder motor control timing	Performance issues	
ARDUINO ONLY	Hard	Arduino, Extra modules	Cheap	Cannot handle "Intelligent" tasks	Not feasible	

Implementation Strategy

To ensure project **organization, realism** and **achievability within the available timeframe**, we have decided to develop the robot in clearly **defined stages**, reducing integration risks, effective individual system tests, and progressive functional expansion.

The subgroups in charge of hardware and system integration must **follow and complete the stage they are in before progressing to the next one** and the software subgroup must **follow along by developing and optimizing the necessary interaction functionalities** according to expectations and requirements.

All requirements and expectations of each stage must be met before introducing any of the optional enhancements, so the core objectives of the project are successfully completed before additional features are considered.

Stage 1: Core Mobility Platform

- **Assembling chassis and tracks** – Stable robot platform;
- **Install motors and configure the motor driver** – Controlled movement;
- **Connect ultrasonic sensors** – Basic obstacle avoidance;
- **Test navigation functions** – Reliable mobility system;

Stage 2: Camera & Interaction System

- **Install trigger button** – User activation input;
- **Install elevated camera mast** – Proper facial capture angle;
- **Configure servo tilt system** – Adjustable camera framing;
- **Connect speakers output** – Audible response system;

Stage 3: Software Integration & Emotion Detection

- **Develop Python control program** – Main robot logic operational;
- **Integrate OpenCV camera capture** – Face detection functional;
- **Add emotion classification system** – Mood recognition active;
- **Establish Pi-Arduino communication** – Full hardware/software integration;

Stage 4: Behaviour Workflow & Testing

- **Test full state workflow** – Smooth user interaction;
- **Improve camera detection timing** – Faster responses;
- **Tune movement reliability** – Stable navigation;
- **Correct software errors** – Reduced failures;

Stage 5: Optional Enhancements

- **Custom voice responses** – More engaging interaction;
- **Voice trigger system** – Hands-free activation;
- **Additional emotions** – Expanded recognition system;
- **Improved robot exterior design** – Stronger visual appeal;

Risks & Considerations

Since this project combines **robotics, electronics, computer vision, and software integration**, several technical and practical risks must be considered during development, because identifying them will allow the team to **apply preventive measures, reduce delays and improve overall reliability** of the final system.

Main Project Risks

RISK CATEGORY	DESCRIPTION	POTENTIAL IMPACT	RECOMMENDED MITIGATION
SYSTEM INTEGRATION	Hardware and software components may fail to communicate correctly	Delays Unstable behaviour	Develop and test each subsystem separately before full integration
CAMERA PERFORMANCE	Poor lighting, incorrect angle, or low image quality may reduce face detection accuracy	Failed emotion analysis	Use adequate lighting, correct camera placement on mast and functional tilt functionality
EMOTION RECOGNITION ACCURACY	Facial expressions may be misclassified by the selected model	Incorrect responses to users	Begin with the recommended small set of clear emotions and improve through testing
POWER MANAGEMENT	Batteries may discharge quickly or provide unstable voltage	Unexpected shutdowns or resets	Use separate power supplies for motors and logic systems
MOVEMENT STABILITY	Uneven surfaces or rapid turning may affect balance	Robot tipping or poor control	Maintain low centre of gravity and test movement speeds
OBSTACLE DETECTION	Sensors may fail to detect nearby objects consistently	Collisions or unsafe movement	Use multiple sensors and adjust detection ranges
TIME CONSTRAINTS	Project scope may exceed available development time	Incomplete final product	Prioritize core features and optional enhancements later
MECHANICAL FAILURE	Loose wiring or weak mounts may fail during operation	Reduced reliability	Secure components properly and inspect regularly

Technical Considerations

CONSIDERATION	IMPORTANCE
SIMPLICITY OF DESIGN	Reduces build time and lowers chance of failure
MODULAR CONSTRUCTION	Easier maintenance and replacement of parts
USER SAFETY	Prevents collisions and protects users during interaction
EASE OF TESTING	Supports faster debugging and progress checks
EXPANDABILITY	Allows future features such as voice control or extra emotions
BATTERY ACCESSIBILITY	Simplifies charging and maintenance

Project Scope Management

One of the most significant risks for student projects is **attempting to implement too many advanced features at the same time**. While functions such as voice commands, advanced navigation, or expanded artificial intelligence may be desirable, these **should only be pursued after** the core system is fully operational.

Overall Risk Assessment

The **overall risk level is considered manageable** if development follows the phased implementation strategy outlined in this report.

By **maintaining realistic scope, testing systems individually**, and **prioritising reliability** over unnecessary complexity, the team can significantly improve the likelihood of successful project delivery.

Conclusion

Based on the alternatives considered, we believe the recommended solution is the best option, proving the optimal balance between **functionality, reliability, cost, and realistic development within the available timeframe.**

Overall, the proposed design is both **achievable** and **innovative**, offering the team a strong opportunity to demonstrate skills in **robotics, programming, and system integration.**

Subject to client approval, the recommendations in this report will be the foundation for the next stages of development.

Appendices

Software Libraries & Development Tools

TOOL	LIBRARY PURPOSE
PYTHON	Main programming language for system logic and “intelligent” functions
OPENCV	Camera access, image processing, and face detection
FER	Pre-trained facial emotion recognition library
DEEPPFACE	Alternative emotion recognition and facial analysis library
PYSERIAL	Communication between Raspberry Pi and Arduino through USB serial connection
ARDUINO IDE	Development environment for Arduino programming
ARDUINO C/C++	Programming language for sensors, motors, and button input
NUMPY	Numerical processing used by Python libraries
GPIO LIBRARIES	Control of Raspberry Pi hardware pins (if required)
VISUAL STUDIO CODE	Recommended code editor for Python development

Note: The final selection of libraries may vary depending on hardware compatibility, system performance, and testing results during development.